SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining 15

Course Faculty : Dr. Terrance Frederick Fernandez

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 10%

QUESTIONS: 5 **Total Marks:25**

Concept Mapping
Annexure A

Criteria	Concept Identification *(2.5)*	Linkages *(2.5)*	Organization *(2)*	Integration of Literature *(2)*	Clarity *(1)*	*Total(10)*
Weightage	25%	25%	20%	20%	10%	100%
Q1	30	24	20	15	10	24
Q2	30	24	20	15	10	24
Q3	30	24	19	15	10	23
Q4	32	24	20	18	10	24
~~Q5~~						

Code of Conduct:

I Sybil Litzy A (192421110) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sybil Litzy A

95/100

Annexure A
Short Answer Test

1.Suppose that a data warehouse consists of the three dimensions time, doctor, and patient, and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit.
(a) Enumerate three classes of schemas that are popularly used for modeling data warehouses.
(b) Draw a schema diagram for the above data warehouse using one of the schema.
(c) To obtain the same list, write an SQL query assuming the data is stored in a relational database with the schema fee (day, month, year, doctor, hospital, patient, count, charge).
(5 Marks)

2.Suppose that a data warehouse for Big University consists of the following four dimensions: student, course, semester, and instructor, and two measures count and avg grade. When at the lowest conceptual level (e.g., for a given student, course, semester, and instructor combination), the avg grade measure stores the actual course grade of the student. At higher conceptual levels, avg grade stores the average grade for the given combination. (5 Marks)
(a) Draw a snowflake schema diagram for the data warehouse.
(b) Starting with the base cuboid [student,course, semester,instructor], what specific OLAP operations
(e.g., roll-up from semester to year) should one perform in order to list the average grade of CS courses
for each Big University student.
(c) If each dimension has five levels (including all), such as "student < major < status < university < all", how many cuboids will this cube contain (including the base and apex cuboids)?
3.Design a data warehouse for a regional weather bureau. The weather bureau has about 1,000 probes, which are scattered throughout various land and ocean locations in the region to collect basic weather data, including air pressure, temperature, and precipitation at each hour. All data are sent to the central station, which has collected such data for over 10 years. Your design should facilitate efficient querying and on-line analytical processing, and derive general weather patterns in multidimensional space. (5 Marks)

4. A data cube, C, has n dimensions, and each dimension has exactly p distinct values in the base cuboid. (5 Marks)
Assume that there are no concept hierarchies associated with the dimensions.
(a) What is the maximum number of cells possible in the base cuboid?
(b) What is the minimum number of cells possible in the base cuboid?
(c) What is the maximum number of cells possible (including both base cells and aggregate cells) in the
data cube, C?
(d) What is the minimum number of cells possible in the data cube, C?

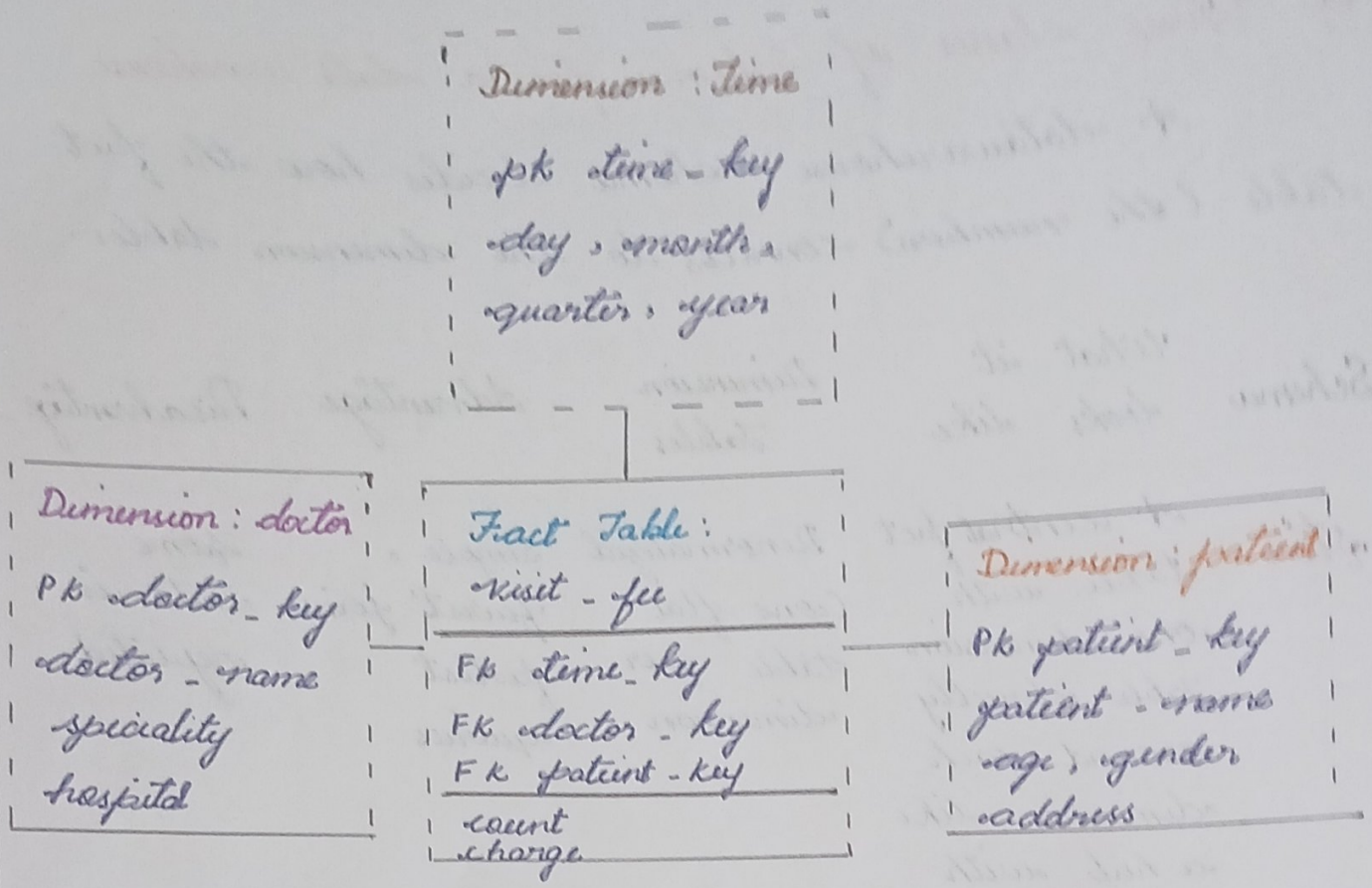
Short Answers Test

1. a) Three classes of Schemas for a data warehouse

A datawarehouse schema decides how the fact table (the numbers) connects to the dimension tables.

Schema	What it looks like	Dimension Tables	Advantage	Disadvantage
Star	A central fact table with one dimension Table directly around each dimension, like a hub with spokes.	Denormalized (one flat table per dimension	Simple, fewest joins, fastest queries	Some data is repeated
Snow Flake	A star whose dimension tables are split further into sub-tables, branching like a snowflake	Normalized	Saves storage and removes redundancy	More joins, slower queries.
Fact Constellation (galaxy)	Several fact tables that share common dimension tables, like cluster of stars	shared between facts	Models several business process together	Most complex to design and maintain

b) Schema diagram : star schemas



Explanation :

* It is a star schema ; one central fact table, visit_fee, joined directly to the three dimension tables (time, doctor, patient)

* holds the foreign keys time_key, doctor_key and patient key and the measures count and charge (fee).

* The dimension attributes are :

-> time : day, month, quarter, year

-> doctor : doctor_name, speciality, hospital

-> patient : patient_name, age, gender, address.

c) SQL query:

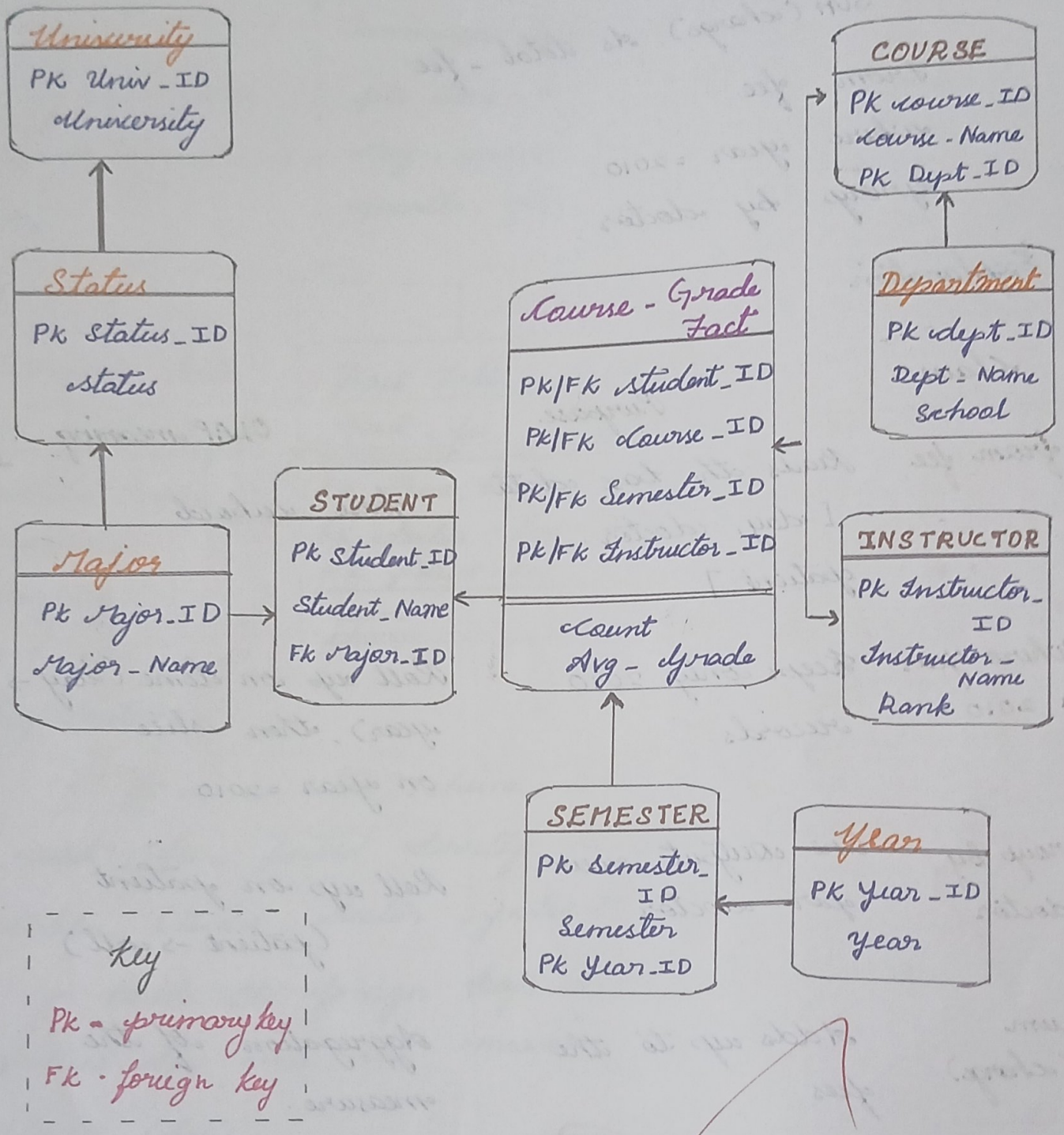
```
SELECT doctor,
    SUM(charge) As total_fee
From fee
where year = 2010
group by doctor;
```

Explanation

Clause	Purpose	OLAP meaning
from fee	Reads the base data [day, doctor, patient]	base cuboid
where year = 2010	Keeps only 2010 records.	Roll up on time (day -> year), then slice on year = 2010
group by doctor	One output row per doctor	Roll up on patient (patient -> all)
sum (charge)	Adds up the fees	Aggregation of the measure.

2. Snowflake Schema for Data warehouse

a)



□ → Dimension Table

□ → Fact Table

□ → Sub dimension Table (Normalized)

2. b) OLAP operations

Goal: The average grade of CS courses of each Big University student.

Step	OLAP operation	Cuboid after the step
1.	Roll-up on course: course_id -> department	[student, department, semester, instructor]
2.	Roll-up on semester: semester -> all	[student, department, all, instructor]
3.	Roll-up on student: student_id -> university	[university, department, all, all]
4.	Dice: University = "Big University" and department = "CS"	Some cuboid, restricted to the selected values
5.	Drill-down on student: university -> student_id (or student_name)	[student, department, all]

c) Number of Cuboids:

There are 4 dimensions and each dimension has 5 levels including ALL.

Therefore:

Number of Cuboids $= 5^4$

$= 5 \times 5 \times 5 \times 5$

$= 625$

Result:

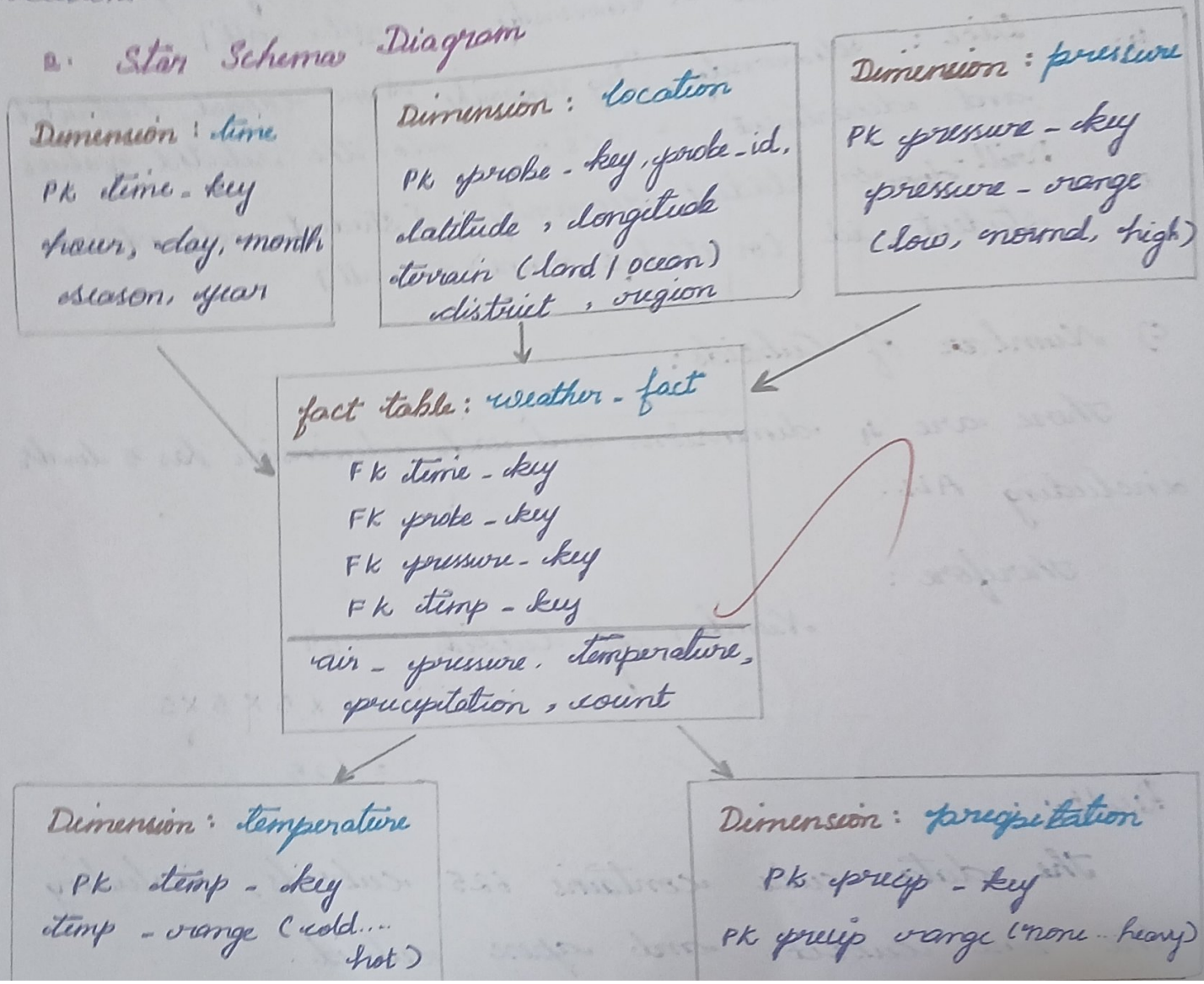
The data cube contains 625 cuboids, including the base cuboid and apex cuboid.

a. Data Warehouse Design for a Regional

1. Proposed Data Warehouse Design:

A multidimensional data warehouse using star schema is suitable for the regional weather bureau because it stores hourly weather measurements collected from approximately 1,000 probes over more than 10 years. It supports efficient querying, OLAP operations and weather pattern analysis across time & geographical locations.

b. Star Schema Diagram



* Loading (ETL): The probes send hourly data to the central station. Clean it (missing or faulty readings), then load it incrementally each hour or day.

* Spatial view: The location dimension has latitude & longitude, so results can be shown as map of regions to show geographic patterns.

6. OLAP operation & patterns

Operation	Example
Roll-up	hour -> day -> month -> season to seasonal averages
Drill-down	region -> district -> probe to find records to extreme values
Slice	month = "July", to view only July data
Dice	terrain = "Ocean" & temp_range = "hot"
Pivot	rotate region against season to compare patterns

7. Sample query

```
SELECT  l.region, t.month,
        AVG(f.temperature) AS avg_temp,
        SUM(f.precipitation) AS total_rain
FROM    weather_fact f
JOIN    time_dim t    ON f.time_key = t.time_key
WHERE   t.year = 2025
GROUP BY l.region, t.month;
```

3. Fact Table and measures

* Fact Table: weather - fact. Its grain is one row per probe per hour
* Measures: air_pressure, temperature, cont & precipitation.

→ At higher levels the measures are aggregated with avg, min, max, sum.

4. Dimensions and concept hierarchies

Dimension	Attributes	Concept hierarchy (low → high)
Time	hour, day, month, season, year	hour < day < month < season < year < all
location	probe - id, latitude, longitude, terrain (land / ocean) region	probe < district < region < all (terrain : land or ocean)
Temperature	temp - range	exact value < range (cold, mild, warm) < all
Precipitation	precip - range	exact value < range (none, light moderate) < all
air pressure	pressure - range	exact value < range (low, normal, high) < all

5. Supporting efficient querying & OLAP

* Indexing: bitmap indexes on the low - cardinality range keys (temperature, precipitation, pressure, terrain) and B-tree index time_key & probe_key.

4. Data cube with n dimensions & p distinct values each

a) Maximum number of cells in the base cuboid

A base cell is one combination of a value from every dimension. Each of the n dimensions has p choices, so the maximum is

$$p^n$$

This happens when every possible combination actually exists in the data.

b) Minimum number of cells in the base cuboid

Each dimension has p distinct values in the base cuboid, so every one of those p values must appear in at least one cell.

One cell can use only one value per dimension, so at least p cells.

$$p$$

c. Maximum number of cells in whole data cube

In any cell of the cube, each dimension is either specific value (p choices) or aggregated into all shown. That gives $p+1$ choices per dimension, so

$$(p+1)^n$$

A cuboid that keeps k dimensions has at most p^k cells. There are $C(n,k)$ such cuboids

$$\sum_{k=0}^{n} \binom{n}{k} p^k = (p+1)^n$$

d) Minimum number of cells in whole data cube

Take the minimum base cuboid from (b), the p diagonal cells. Count the cells in every cuboid. There are 2^n cuboids in all

Cuboids	How many	Cells in each
Base cuboid (all n dimension)	1	P
Apex cuboid (all dimension aggregated to *)	1	1
Every other cuboid (at least one dimension kept)	$2^n - 2$	P

$$\text{Total} = p + (2^n - 2)\,P + 1 = (2^n - 1)P + 1$$

Numerical example ($n = 3, p = 4$)

Part	Formula	Value
a) Max base cells	p^n	$4^3 = 64$
b) Min base cells	p	4
c) Max cell in cube	$(p+1)^n$	$5^3 = 125$
d) Min cells in cube	$(2^n - 1)P + 1$	$7 \times 4 + 1 = 29$